Fig. 4: Synchronising switching along a pair of lines (Credit: Columbia University)

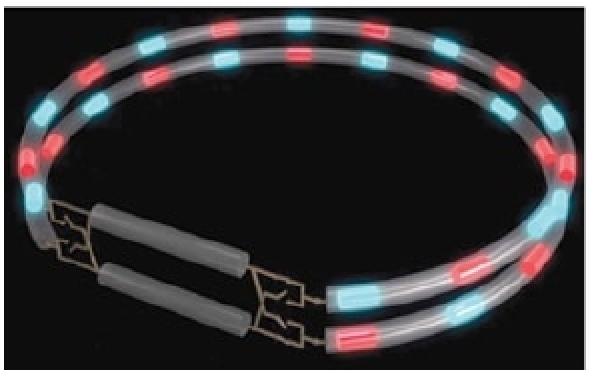

Fig. 5: Non-staggered communicator-based device (gyrator) (Credit: Columbia University)

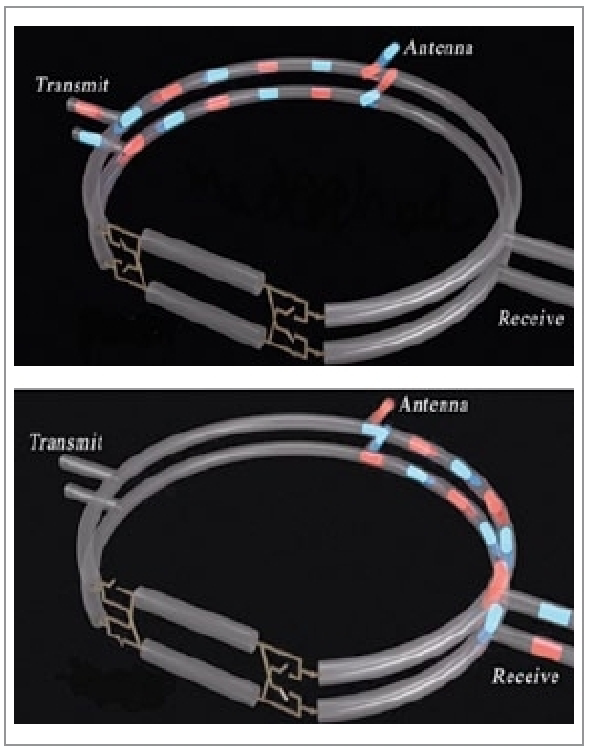

Fig. 6: Circulator realised on non-staggered communicator (Credit: Columbia University)

erties of transistors could be used. However, this device has never been commercialised, because it suffers from fundamentally poor noise cancellation. This long-held assumption has resulted mainly from the need for isolation between transmitter and receiver nodes. This is also known as self-interference cancellation, which needs to be greater than 100db.

Despite significant research efforts in radio frequency (RF) domain, non-reciprocal components based on magnetic materials remain incompatible with complementary metal oxide semiconductor (CMOS) IC fabrication processes. This is due to material incompatibilities and the need for a magnetic field bias, which significantly restricts their impact.

Fig. 7: Circulator IC compared to traditional circulator (Credit: Columbia University)

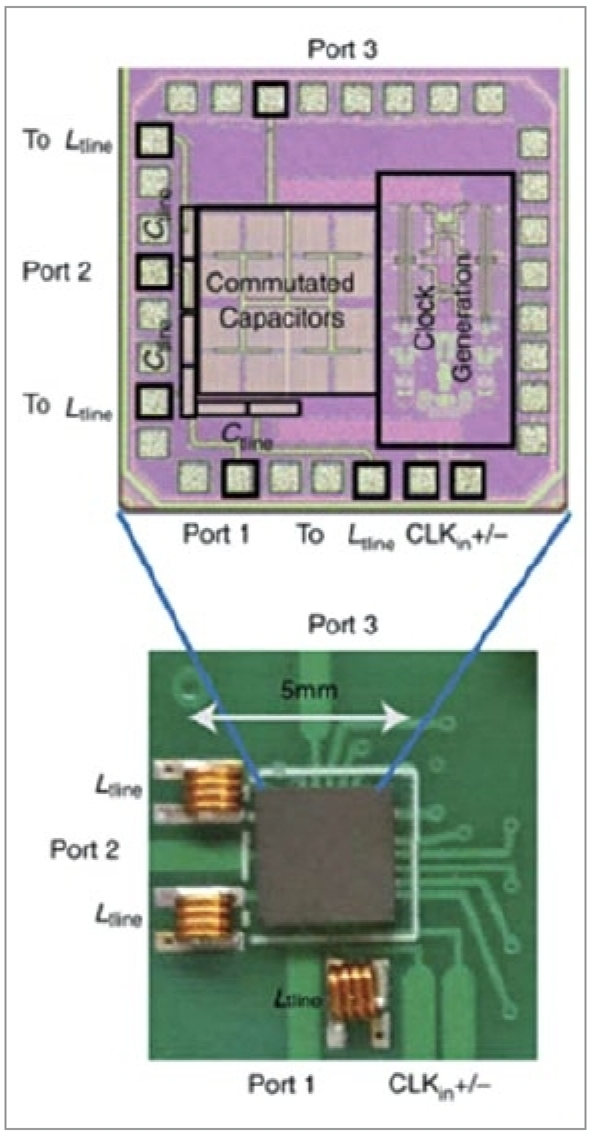

Fig. 8: Circulator IC embedded into communication circuit (Credit: Columbia University)

Recently, several groups have presented a new class of magnetic-free non-reciprocal devices based on linear periodically time variant circuits. These can overcome the limitation of all previous approaches.

Harish Krishnaswamy, associate professor at Columbia University, USA, has devised a way of providing a full duplex solution that can also provide the first critical 20dB - 30dB of the total self-interference cancellation (SIC). Magnetic-free, linear and passive phase non-reciprocity based on staggered communication and a highly-miniaturised RF circulator has been introduced. It embeds the phase non-reciprocal component with a ring resonator. The non-staggered communicator network does this by carefully synchronising switching along a pair of lines where reciprocity can be broken. Delay of line is one-quarter of the switching period [(τ = T/4)].

Switches at the right are delayed with respect to switches on the left, by quarter of period and toggled between straight and criss-cross connections. So, when the signal entering from the left side reaches the right, the switch is toggled to straight connection, and the signal neatly exits the path.

However, when the signal entering from the right side reaches the left, the switch is toggled to the criss-cross connection, and the signal experiences suppression.

Based on the above concept, a CMOS IC has been developed, having dimensions smaller than a penny. It can be integrated into circuits and, hence, uses a single antenna for both transmission and reception of signals on a single channel at the same time.

Prof. Krishnaswamy explains, "Full duplex communication—where transmitter and receiver operate at the same time and at the same frequency—has become a critical research area. It has been shown that Wi-Fi capacity can be doubled on a nano-scale silicon chip with a single antenna. This has enormous implications for devices like smartphones and tablets."

Co-researcher, Jin Zhou, adds, "Being able to put the circulator on the same chip as the rest of the radio can significantly reduce the size of the system, enhance its performance and introduce new functionalities critical to full duplex."

In 2017, Prof. Krishnaswamy co-founded MixComm Inc., a venture-backed startup, to commercialise CoSMIC Laboratory's advanced